# Jingyu WU

1512 Dempster Street Unit3, Evanston, IL 60202, USA

## Education

**Northwestern University** — Evanston, IL

*Master of Science in Computer Science (GPA: 3.9/4.0)* — *Sep. 2023 – Mar. 2025(Expected)*

**Coursework:** Deep Learning for NLP, Rapid Prototyping for Software Innovation, Database, Computer Networking, etc.

Served as Peer Mentor of CS449 Deep Learning

**Hong Kong University of Science and Technology (HKUST)** — Hong Kong

*Bachelor of Science in Data Science (Graduate with Dean's list)* — *Aug. 2019 – May. 2023*

**Coursework:** Object Orienting Programming, Data Structure and Algorithm, Deep Learning, Machine Learning, C++, Python

**Honors:** 2021/22 HKSAR Government Fund Overseas Expansion Award; 2021/22 HKUST Study Abroad Scholarship

## Tech Stack

**Language**: Python, C++, JavaScript/TypeScript, C, SQL, HTML, Bash/Shell, Latex, R
**Developer Tools**: AWS Cloud Service, Git, Figma, Docker, Postman, Firebase, Jupyter Notebook, Anaconda, Trello
**Database**: PostgreSQL, MySQL
**Web Technologies**: ReactJS, Node.js, RESTful API, HTTP

## Professional Experience

**ByteDance (TikTok)** — June 2024 – present

*Software Engineer Intern* — *Hybrid*

- Researched, deployed, and optimized **LLM**s with Llama.cpp and mlc-mnn on PC and Android, leveraging **quantization** techniques; actively contributed to the **Llama.cpp** open source community.
- Developed and deployed a Kotlin-based chatbot on **Android**, testing 3+ models on 15+ devices.
- **Finetuned** models for intent recognition (90%, accuracy) and text summarization, serving 100M+ users.
- Converted custom attention-based **PyTorch** model to **ONNX** and researched deployment strategies.

**EXA App Limited** — June 2022 – Feb. 2023

*Software Engineer Intern* — *Hong Kong Science Park*

- Designed and developed a robust and scalable backend system from scratch using **TypeScript**, **Node.js**, **ExpressJS**, **Knex**, and **PostgreSQL** to support both web and mobile applications for a medical institution.
- Implemented an **authentication and authorization** system to efficiently handle user roles and permissions.
- Optimized the data comparison algorithm, achieved 90% increase in query speed performance.
- Communicated with client and extended API to improve user experience.

## Selected Projects

**XTrips.ai: An intelligent flight booking system** — Sep. 2023 – Jan. 2024

- Designed JSON based **RESTful APIs** using AWS API Gateway and tested with **Postman**, implemented the lambda functions with JavaScript.
- Architected a robust backend and APIs using **Express.js** hosted on AWS **EC2**, managing user authentication, flight queries, bookings, and personalized travel itinerary generation.
- Integrated **AWS RDS** (**MySQL**) for reliable data management and AWS **S3** for secure storage of itinerary files.
- Implemented server-client structure for seamless flight booking, applied **LLM** for itinerary recommendation.

**FurNitUre Web Application - App Link** — Sep. 2023 – Jan. 2024

- Prototyped a single-page web application using **ReactJS** for NU students for second-hand furniture exchange.
- Accomplished the decoding and visualization of geo-location information using Geocoding and Google Map API.
- Integrated user authentication with email validation using EmailJS, styled the components with **CSS**.

**Database Management System Built From Scratch** — Sep. 2023 – Dec. 2023

- Developed a thread-safe buffer pool manager with **C++**, implemented comprehensive unit test with **GTest**.
- Implemented the LRU-K eviction algorithm to manage buffer pool memory, optimizing page retrieval and storage.
- Built a disk scheduler to handle asynchronous read/write operations to disk, enhancing data access efficiency and reducing I/O bottlenecks.

**Deep Learning Library Automation Testing Tool - Github Link** — June 2(

- Collaborated with two teammates to build a testing tool for deep learning libraries (PyTorch, Tensor customizable parameter testing.
- Designed and implemented an end-to-end pipeline in **Python** for generating, storing, and comparing results, with robust error handling and logging mechanisms.
- Conducted experiments and analyzed results using **Matplotlib**,**NumPy**.